

The Hippocampus

Part 1: Anatomy and Overview

Computational Neuroscience

University of Bristol

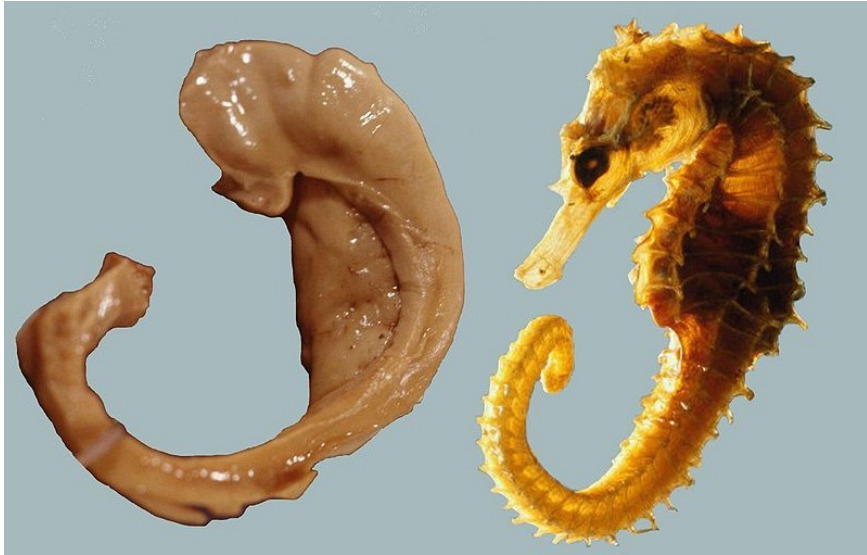
mrule

Learning outcomes:

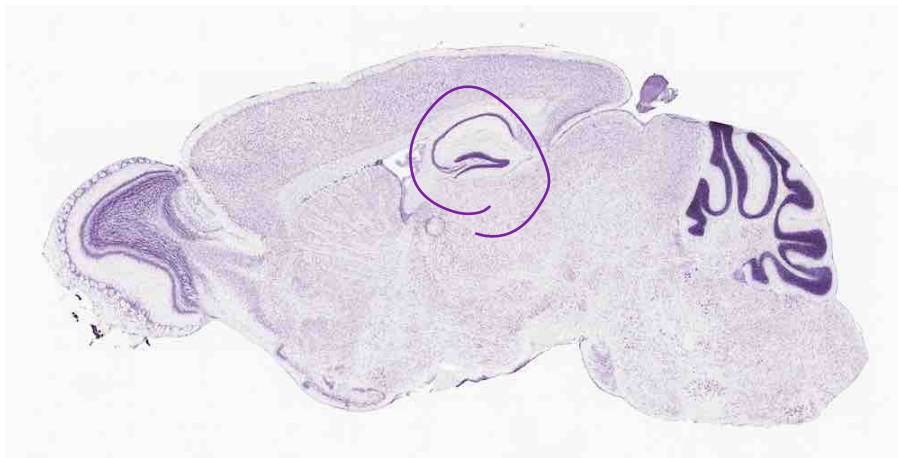
- ▶ Be able to describe basic anatomical location and features of the hippocampus
- ▶ Be able to name two key cognitive functions of the hippocampus
- ▶ Become familiar with hippocampal regions we will study in the context of memory and navigation

Hippocampus = (horse + sea monster)

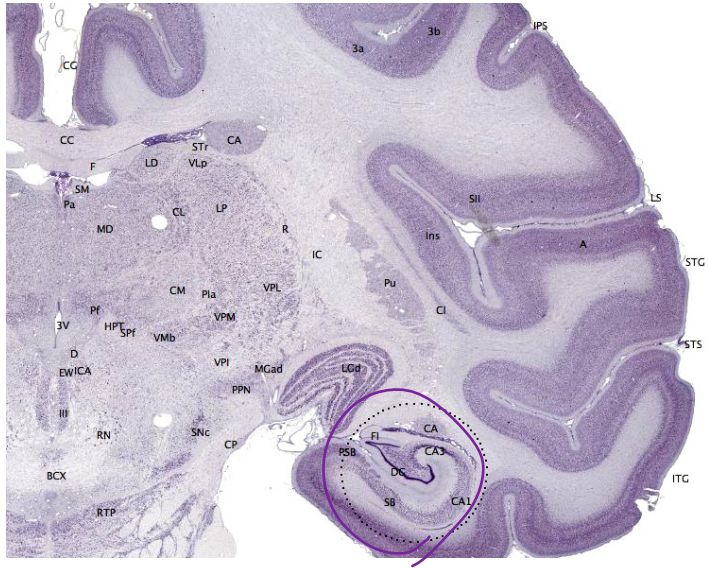
CA = Cornu Ammonis



[L Seress' dissection of hippocampus and fornix, and a seahorse](#)

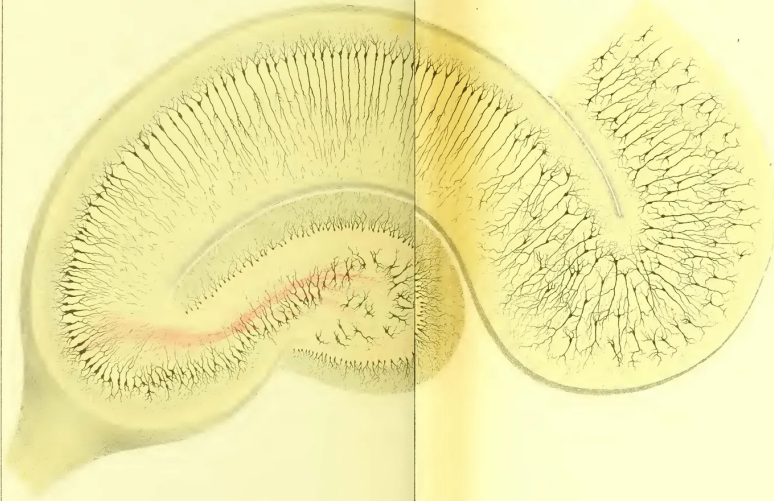


<https://mouse.brain-map.org>



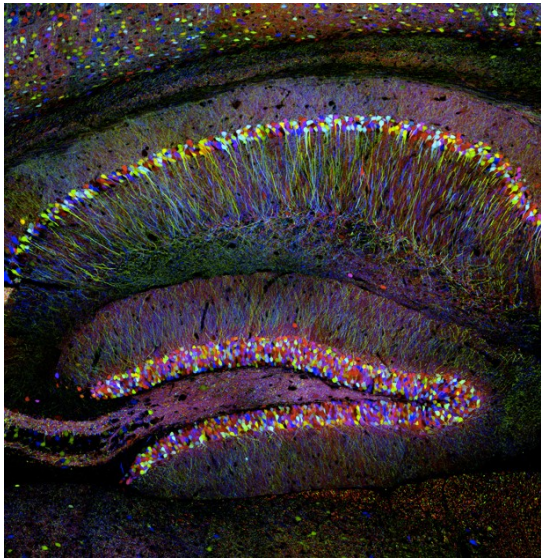
brainmaps.org

Circle the hippocampus



Hippocampus, Camillo Golgi

Label the parts and say what they do



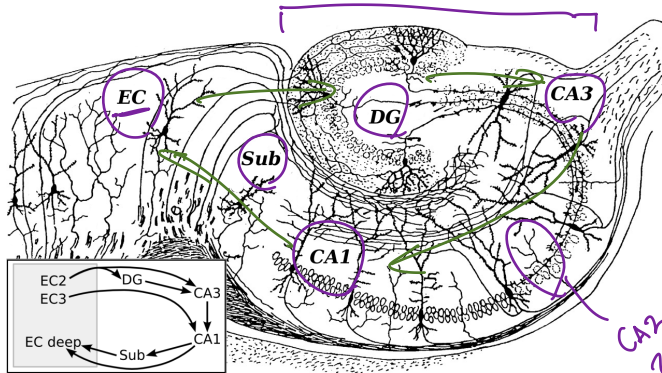
"Brainbow", Livet & al. (2007) Nature

Hippocampus

Archicortex, deep inside temporal lobe

You need it to **form new memories** and **navigate the world**

HIPPOCAMPUS Hippocampal formation.



Entorhinal Cortex (EC)

Dentate Gyrus (DG)

Cornu Ammonis field 3 (CA3)

Cornu Ammonis field 1 (CA3)

Subiculum (Sub)

Modified from Cajal (1911).

We will discuss these more in context of *memory* and *navigation*

The tri-synaptic loop

Entorhinal cortex (EC): Gateway

(MEMORY)

Trisynaptic loop

- ▶ Dentate Gyrus (DG) → CA3 → CA1 → Subiculum

CA3 the *only* subregion with *substantial recurrent excitatory connectivity*

- ▶ recurrent attractor network
- ▶ pattern completion

end

